

III B. Tech II Semester Regular Examinations, JULY-2023
OBJECT ORIENTED ANALYSIS AND DESIGN
 (Com. To CSE & IT)

Time: 3 hours

Max. Marks: 70

Answer any **FIVE** Questions **ONE** Question from **Each unit**
 All Questions Carry Equal Marks

UNIT-I

1. a) Why software is inherently complex? Explain in detail. [7M]
 b) What are requirements for the satellite navigation system? Explain. [7M]
 (OR)
2. a) Explain about the meaning of design. Describe the steps in designing process. [7M]
 b) Discuss about validating the proposed system architecture of satellite navigation system. [7M]

UNIT-II

3. a) Describe common modeling techniques of relationships. [7M]
 b) Explain about determining system use cases of the train traffic management system. [7M]
 (OR)
4. a) Briefly discuss about the conceptual model of the UML. [7M]
 b) Discuss about defining the TTMS architecture. [7M]

UNIT-III

5. a) How to identify classes in class diagrams? Explain. [7M]
 b) Discuss about the elaboration phase of Cryptanalysis. [7M]
 (OR)
6. a) Describe common modeling techniques of Packages. [7M]
 b) What are requirements of Cryptanalysis? Explain about analysis of knowledge sources. [7M]

UNIT-IV

7. a) What are interaction diagrams? Explain with examples. [7M]
 b) Describe client and implementation views of Vacation Tracking System. [7M]
 (OR)
8. a) Explain common modeling techniques of activity diagrams. [7M]
 b) Describe service data objects and primary key generation of Vacation Tracking System. [7M]

UNIT-V

9. a) "State machines are used for behavioral modeling". . Justify. [7M]
 b) Discuss about scenarios of Weather Monitoring System. [7M]
 (OR)
10. a) Describe deployment diagrams with suitable examples. [7M]
 b) Discuss about requirements and post transition of Weather Monitoring System. [7M]



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UNIT-I

1. a) What are categories of analysis and design methods? Explain. [7M]
- b) Explain about allocating nonfunctional requirements and specifying interfaces in satellite-based navigation. [7M]

(OR)

2. a) Discuss about the canonical form of a complex system. [7M]
- b) How to determine mission use cases of satellite-based navigation? Explain [7M]

UNIT-II

3. a) Discuss various phases of software development life cycle. [7M]
- b) Explain about Key Abstractions and Mechanisms of TTMS. [7M]

(OR)

4. a) Explain common modeling techniques of classes. [7M]
- b) Describe train schedule planning of TTMS. [7M]

UNIT-III

5. a) What is an object diagram? Explain identifying objects in object diagrams. [7M]
- b) Discuss about designing the controller of Cryptanalysis. [7M]

(OR)

6. a) What are advanced relationships in UML? Explain. [7M]
- b) Describe dependencies and affirmations of Cryptanalysis. [7M]

UNIT-IV

7. a) Explain the importance use case diagram with example. [7M]
- b) Discuss about use case view of Vacation Tracking System. [7M]

(OR)

8. a) Describe terms and notations of use cases with suitable examples.. [7M]
- b) Explain the inception phase of Vacation Tracking System. [7M]

UNIT-V

9. a) What are common modeling techniques of Component diagrams? Explain. [7M]
- b) Describe sensor mechanism and display mechanism of Weather Monitoring System. [7M]

(OR)

10. a) How events and signals are useful in UML? Explain. [7M]
- b) Discuss about use cases of Weather Monitoring System. [7M]



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UNIT-I

1. a) What are the five attributes of a complex system? Explain. [7M]
b) Draw and explain the white-box activity diagram for control launch of satellite-based navigation [7M]

(OR)

2. a) Explain about arithmetic and object-oriented decompositions. [7M]
b) Describe the satellite navigation system's post transition. [7M]

UNIT-II

3. a) What are basic building blocks of the UML? Explain. [7M]
b) Discuss about Subsystem Specification of TTMS. [7M]

(OR)

4. a) Explain about classification of UML diagrams in detail. [7M]
b) Describe message passing in TTMS with neat diagrams. [7M]

UNIT-III

5. a) What are common modeling techniques of advanced relationships? Explain. [7M]
b) Describe designing the knowledge sources of Cryptanalysis. [7M]

(OR)

6. a) Write and explain terms and concepts of interfaces, types and roles. [7M]
b) Discuss about post transition of Cryptanalysis. [7M]

UNIT-IV

7. a) Explain about terms and concepts of activity diagrams. [7M]
b) Describe entities of Vacation Tracking System. [7M]

(OR)

8. a) What are common modeling techniques of interactions? Explain. [7M]
b) Explain the web pages and the user interface of Vacation Tracking System. [7M]

UNIT-V

9. a) Discuss about state chart diagrams in detail. [7M]
b) Discuss about the frame mechanism of Weather Monitoring System. [7M]

(OR)

10. a) Describe common modeling techniques of deployment diagrams. [7M]
b) What are Weather Monitoring System Use Cases? Explain. [7M]



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UNIT-I

1. a) Discuss about the roles of abstraction and hierarchy in detail. [7M]
- b) Describe validating the proposed system architecture of satellite-based navigation [7M]

(OR)

2. a) Explain about the structure of complex systems. [7M]
- b) Describe stipulating the system architecture and its deployment. [7M]

UNIT-II

3. a) What is modeling? Explain the importance of modeling. [7M]
- b) Discuss about analyzing system functionality of TTMS. [7M]

(OR)

4. a) Explain about relationships in the UML with examples. [7M]
- b) What are requirements for the Train Traffic Management System? Explain. [7M]

UNIT-III

5. a) Describe terms and concepts of advanced classes. [7M]
- b) Briefly explain about inception phase of Cryptanalysis. [7M]

(OR)

6. a) Explain common modeling techniques of class and object diagrams. [9M]
- b) Describe blackboard objects of Cryptanalysis. [5M]

UNIT-IV

7. a) Explain about common modeling techniques of Use cases. [7M]
- b) Discuss about analysis and design model of Vacation Tracking System. [7M]

(OR)

8. a) Describe activity diagrams for School Management System. [7M]
- b) Explain logical and process views of Vacation Tracking System. [7M]

UNIT-V

9. a) Discuss about terms and concepts of threads, time and space. [7M]
- b) Explain about defining the Problem Boundaries of Weather Monitoring System. [7M]

(OR)

10. a) Explain about component diagrams in detail. [7M]
- b) Describe the user interface mechanism of Weather Monitoring System. [7M]

